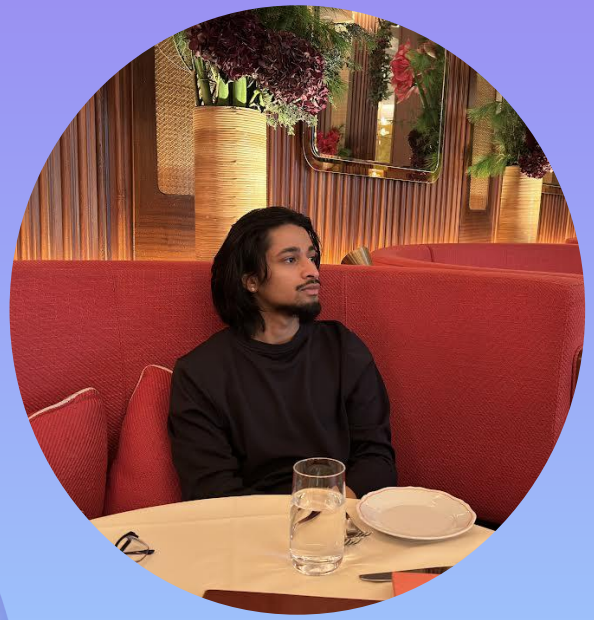
INDEPENDENT STUDY IN C.S. [PROPOSAL]

Rahul Sankaralingam

ABOUT ME



Rahul Sankaralingam

-Senior studying B.S. in Computer Science at Rutgers University
-Transferred from Rensselaer Polytechnic Institute in 2022
-Have a deep passion for innovation and modelling in the tech/music industry
-Website: https://rulah.me

PROPOSAL OVERVIEW

- Analyzing Cape May, NJ's housing market
- Evaluating ROI on improvements
- Using datasets (e.g. Zillow, Realtor.com) for market analysis
- Studying the impact of demographic changes and amenities
- Developing predictive models
- Implementing Python and modelling the neighborhood
- Using machine learning concepts to predict real estate market trends

COURSE OVERVIEW

- Real estate market analysis using CS principles
- Using predictive modeling
- Exploring machine learning trends
- Hands-on experience with real estate data and market prediction
- Analyze practical market insights
- Opportunity to translate CS education into analysis skills

C.S. EXPERIENCE

- 01:198:170 - Computer Application for Business
- 01:198:112 - Data Structures
- 01:198:336 - Principles of Information and Data Management
- Enrolled in 01:198:211 - Computer Architecture
- https://rulah.me
- HTML/CSS
- Backend development and user data management as Python Developer Intern at Entrans Inc.
- Engaged in video game development and music production projects

PROJECT IMPLEMENTATION

- Developing interactive neighborhood visualization for a property in Cape May, similar to Apple Maps, through Python
- Incorporating clickable features for each property and amenity (schools, stores, etc.)
- Using data visualization tools to present information through charts, graphs, and regression lines
- Provide neighborhood insights with interactive data exploration features
- Focusing on user-friendly design for effective real estate market analysis

MENTORSHIP

- I am seeking your mentorship and guidance into computer science applications in real-world scenarios
- Interested to know more about the housing market and apply it in the future
- Aiming to apply Rutgers CS curriculum knowledge to practical business technology challenges
- Schedule a meeting once a week (Fridays or Saturdays?)
- Propose rough drafts weekly, with the final due by 5/2/24

CONCLUSION

- Bridging academic knowledge with housing models
- Requesting the project to be considered for 4 elective credits in the CS curriculum
- Expressing a strong desire for practical experience in data-driven and real-world applications and utilizing this information in the future!

THANK YOU!